

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Q1. Remote Rural Branch Office

1. Problem Understanding and Constraints

The company is establishing a branch office in a **remote rural location where high-speed wired Internet is unavailable**. The network must support **cloud-based applications, video conferencing, and secure communication with headquarters**. The major constraints are:

- Limited budget for connectivity
- Unavailability of high-speed wired Internet
- Requirement for uninterrupted business operations
- Need for sufficient bandwidth for cloud applications and video conferencing
- Requirement for secure communication with headquarters
- Need for future scalability

2. Problem Decomposition

The problem can be divided into four major areas:

1. **Internet connectivity** – selecting an appropriate connection for the rural location.
2. **Network infrastructure** – selecting routers, switches and wireless devices.
3. **Security** – protecting communication between the branch and headquarters.
4. **Reliability and scalability** – ensuring continuous operation and allowing future expansion.

3. Proposed Solution

Internet Connection:

Use a **4G/5G fixed wireless connection** as the primary Internet connection where cellular coverage is available. If reliable cellular service is unavailable, a **satellite Internet connection** can be used as an alternative.

Networking Devices:

- Enterprise-grade router with dual-WAN capability
- Managed Gigabit Ethernet switch
- Wireless access points for employees
- UPS for networking equipment
- 4G/5G modem or satellite terminal

Security Mechanisms:

- Site-to-site **IPsec VPN** between the branch and headquarters
- Firewall integrated into the router

- WPA3-secured wireless access
- Strong user authentication
- Network segmentation using VLANs
- Regular firmware and security updates

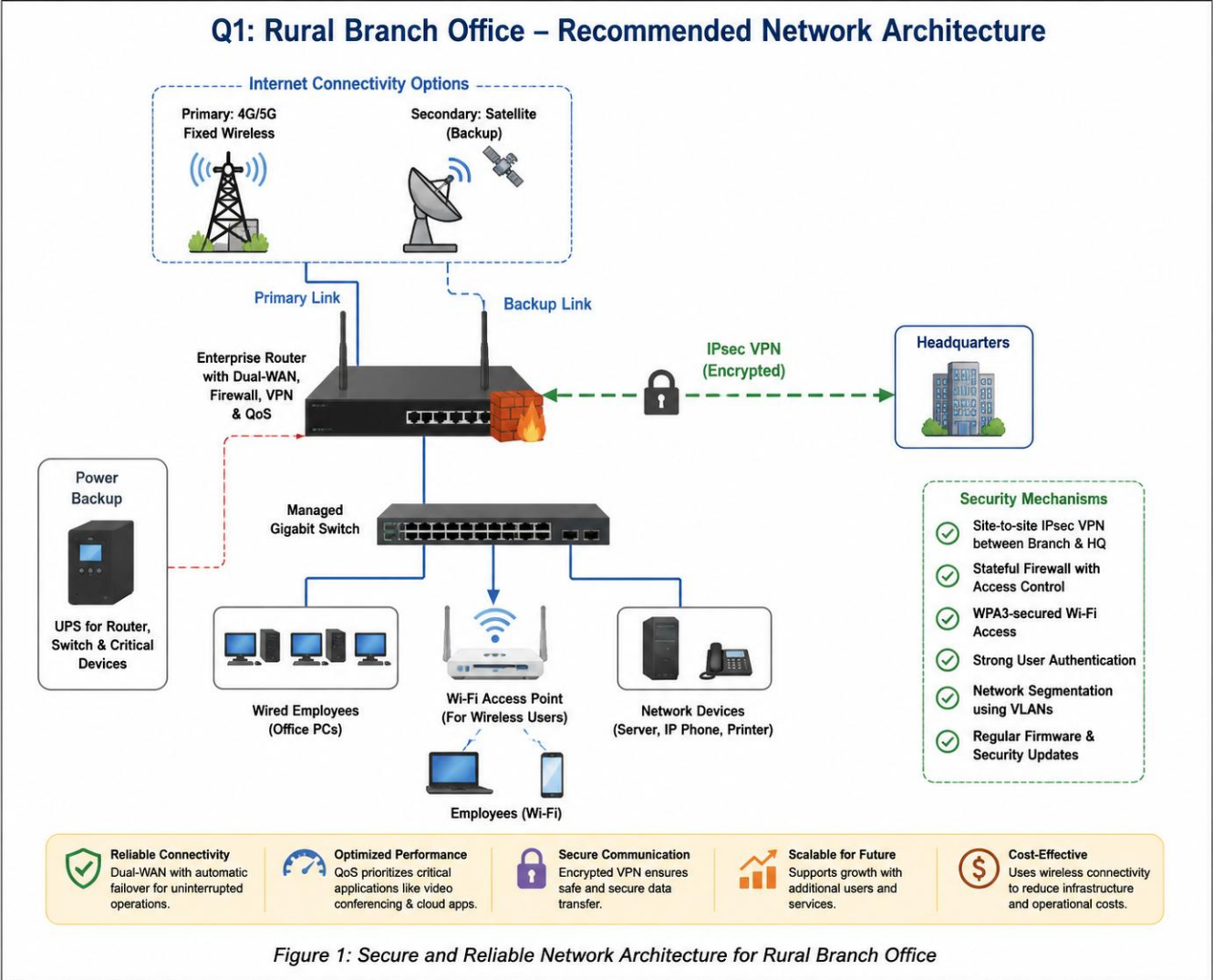


Figure 1: Secure and Reliable Network Architecture for Rural Branch Office

4. Application of Constraints

Constraint	Proposed Solution
Rural location	4G/5G or satellite connectivity
Limited budget	Use wireless connectivity instead of expensive wired infrastructure
Video conferencing	Prioritize real-time traffic using QoS
Cloud applications	Adequate bandwidth and traffic prioritization
Secure headquarters communication	Site-to-site IPsec VPN
Uninterrupted operation	Dual-WAN + UPS
Future growth	Scalable router, managed switch and VLAN

	architecture
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5. Logical Reasoning and Strategy

A wireless primary connection avoids the high installation cost associated with extending wired infrastructure to a rural location. **QoS** can prioritize video conferencing and business-critical cloud traffic. A **VPN** provides encrypted communication between the branch and headquarters. For improved reliability, the router can use **dual-WAN failover**, allowing a secondary connection to take over when the primary connection fails.

6. Feasibility and Innovation

The proposed design is feasible because it uses technologies that can operate without a wired broadband infrastructure. The combination of **wireless Internet + VPN + QoS + dual-WAN failover** provides a balance between cost, performance, security and reliability. An innovative aspect is using **automatic WAN failover and traffic prioritization**, which maintains critical business services even during connectivity problems.

7. Conclusion

Therefore, a **4G/5G-based network with satellite/secondary connectivity, a dual-WAN router, firewall, managed switch, VPN, VLANs, QoS and UPS** is a suitable solution. It satisfies the constraints of **bandwidth, reliability, cost, security and future scalability** while supporting uninterrupted business operations.

Q2. University Wireless Network

1. Problem Understanding and Constraints

The university needs campus-wide wireless connectivity for **more than 3,000 students** across classrooms, laboratories, libraries and hostels. The major constraints are:

- More than 3,000 simultaneous users
- Limited number of wireless access points
- Large physical coverage area
- High traffic demand
- Signal interference must be minimized
- Secure user access is required
- Budget is limited
- Network must balance coverage and performance

2. Problem Decomposition

The problem can be divided into:

1. **Coverage planning** – determining strategic AP locations.
2. **Capacity planning** – supporting large numbers of simultaneous users.
3. **Frequency planning** – minimizing interference.
4. **Security** – authenticating students and protecting network access.
5. **Cost optimization** – achieving maximum coverage with limited APs.

3. Proposed Wireless Design

Use centrally managed **Wi-Fi 6/6E access points** with a wireless controller or cloud-based management system.

Access Point Placement

APs should be positioned based on **user density rather than simply geographical distance**.

Priority areas:

- Classrooms – APs near high-density student areas
- Laboratories – dedicated APs where device density is high
- Libraries – additional capacity due to simultaneous users
- Hostels – APs distributed across floors/buildings

Avoid placing APs behind thick walls, metal structures or other obstacles.

4. Frequency Band Selection

Use:

- **5 GHz** as the primary band for high-performance users.
- **2.4 GHz** mainly for legacy/IoT devices and better long-distance coverage.
- **6 GHz** where supported and permitted, for high-capacity compatible devices.

Use proper channel planning to minimize co-channel and adjacent-channel interference.

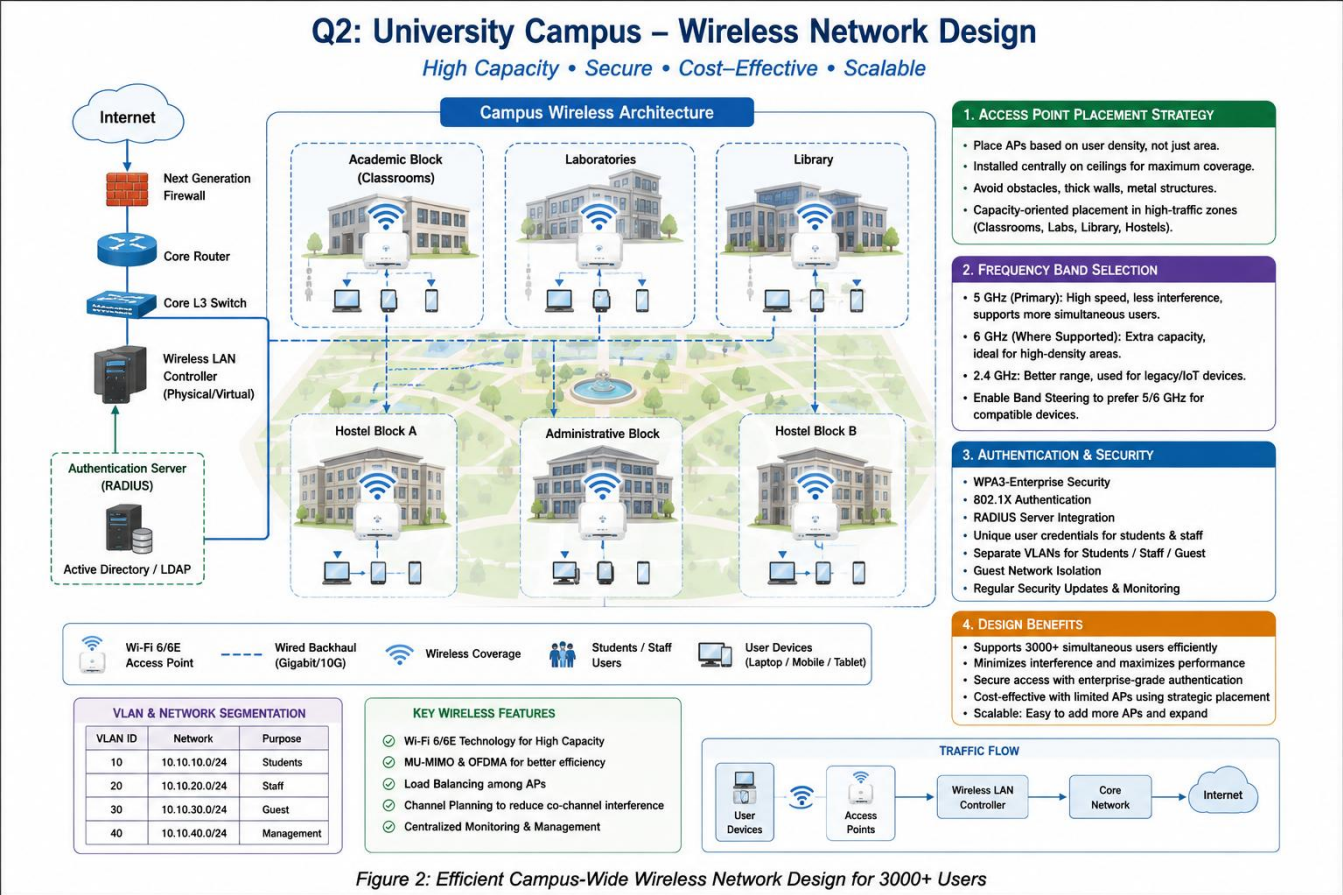


Figure 2: Efficient Campus-Wide Wireless Network Design for 3000+ Users

5. Authentication and Security

Implement:

- **WPA3-Enterprise**
- 802.1X authentication
- RADIUS authentication server
- Individual student credentials
- Separate staff/student networks
- VLAN-based segmentation
- Guest network isolation

6. Application of Constraints

Constraint	Solution
3,000+ users	Wi-Fi 6/6E and high-capacity APs
Limited AP count	Strategic high-density placement
Interference	Channel planning and band steering

Security	WPA3-Enterprise + 802.1X
Budget	Prioritize high-density areas
Coverage	Proper AP placement and RF planning
Scalability	Centralized management and expandable architecture

7. Logical Strategy

Instead of installing APs uniformly across the campus, AP deployment should be based on **user density and traffic requirements**. This allows the university to achieve better performance with fewer APs. Band steering can encourage compatible devices to use higher-capacity bands, while channel planning reduces interference.

8. Creativity and Innovation

A **centralized intelligent wireless management system** can monitor AP utilization, interference and connected users. APs can then be optimized based on actual usage patterns.

9. Conclusion

The proposed **Wi-Fi 6/6E centrally managed architecture with strategic AP placement, RF/channel planning, WPA3-Enterprise and 802.1X authentication** provides a cost-effective balance between **coverage, capacity, security, performance and scalability**.

Q3. Financial Institution Secure Network

1. Problem Understanding and Constraints

The financial institution requires a secure internal network that allows employees to access **sensitive customer information and online banking services** while maintaining network performance.

The major constraints are:

- Strong cybersecurity
- Protection of sensitive customer information
- Secure online banking access
- Compliance with strict security policies
- Minimal impact on network performance
- Controlled operational cost
- Reliable authentication
- Detection of malicious activity

2. Problem Decomposition

The solution can be divided into:

1. **Network segmentation**
2. **Firewall deployment**
3. **Authentication and access control**
4. **Intrusion detection**
5. **Secure Internet and banking access**
6. **Monitoring and compliance**

3. Proposed Network Architecture

A **multi-layer segmented architecture** should be implemented.

Example:

Internet → Edge Firewall → DMZ → Internal Firewall → VLANs → Secure Servers

Create separate VLANs for:

- Employee systems
- Customer-data systems
- Banking/application servers
- Management systems

- Guest users

Sensitive customer-data systems should not be directly accessible from ordinary employee networks.

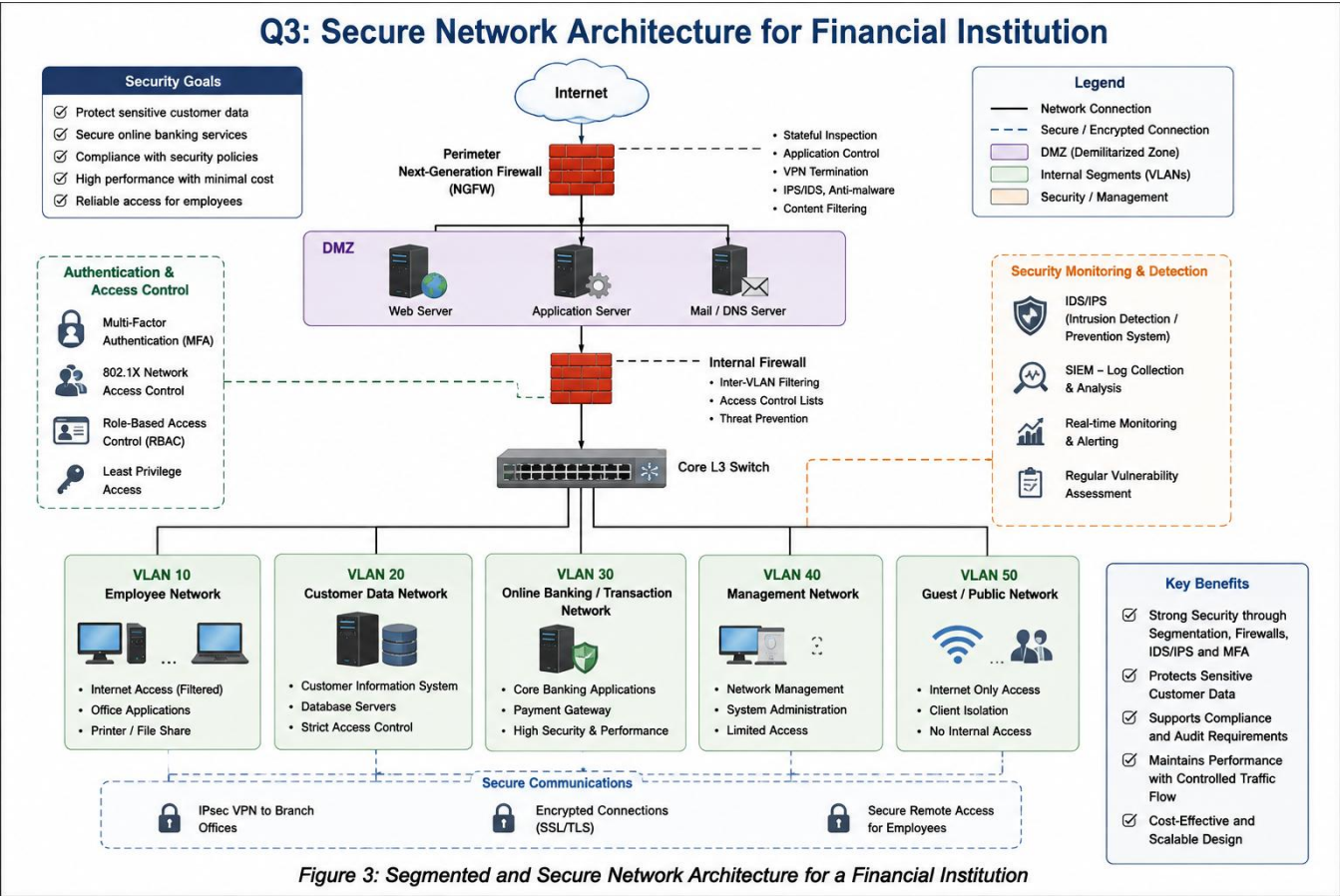


Figure 3: Segmented and Secure Network Architecture for a Financial Institution

4. Firewall Deployment

Use a **next-generation firewall (NGFW)** at the network perimeter.

The firewall should provide:

- Stateful packet inspection
- Application filtering
- Access-control policies
- VPN support
- Intrusion prevention
- Logging and monitoring

A separate internal firewall can further restrict communication between sensitive network segments.

5. Authentication

Use:

- Multi-factor authentication (MFA)
- 802.1X network authentication
- Role-based access control
- Strong passwords
- Least-privilege access

Employees should receive access according to their job responsibilities.

6. Intrusion Detection

Deploy **IDS/IPS** to monitor network traffic and detect suspicious activities.

Security monitoring should identify:

- Unauthorized access attempts
- Malware activity
- Port scanning
- Abnormal traffic
- Suspicious login behavior

Logs should be centrally collected for monitoring and auditing.

7. Application of Constraints

Constraint	Solution
Sensitive customer information	VLAN segmentation + strict ACLs
Cybersecurity	NGFW + IDS/IPS
Compliance	Logging, monitoring and access control
Performance	Proper firewall sizing and traffic policies
Operational cost	Centralized security management
Secure employee access	MFA + RBAC
Online banking security	DMZ + firewall policies + encrypted connections

8. Logical Reasoning and Strategy

Network segmentation limits the impact of a security breach. If an attacker compromises an employee workstation, VLAN isolation and firewall rules prevent direct access to sensitive customer databases.

MFA reduces the risk associated with stolen credentials, while IDS/IPS provides continuous monitoring for suspicious activity.

The architecture therefore follows a **defense-in-depth strategy**, where multiple security controls work together rather than depending on a single security mechanism.

9. Creativity and Innovation

A **zero-trust approach** can be incorporated so that users and devices are continuously verified before accessing sensitive resources. Security policies can also be adjusted according to user role, device status and requested resource.

10. Conclusion

The recommended architecture combines **network segmentation, VLANs, NGFW, DMZ, MFA, RBAC, IDS/IPS and continuous monitoring**. This provides strong protection for sensitive financial information while maintaining performance, meeting compliance requirements and controlling operational costs.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand